

Relatório de Projeto Web

Relatório técnico-académico

Documento académico com registos textuais e visuais de commits, branches, merge com conflito, tags e preparação de colaboração GitHub.

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Tema	Git, GitHub, branches, merge, tags e registos visuais

Resumo

O trabalho demonstra a criação, evolução e análise de um repositório Git com foco em rigor operacional, documentação de registos e preparação para colaboração no GitHub. A estrutura inclui comandos essenciais, análise do histórico, resolução de conflito de merge, tagging semântica e material visual de suporte.

Palavras-chave

Git, GitHub, branches, merge, conflito, tags, documentação, registo visual

Registos principais	Screenshots SVG, report.md e logs de Git
Colaborador	vicsolucoes
Estado	Relatório revisto e pronto para submissão

Registos Visuais

As figuras seguintes sintetizam os momentos-chave do trabalho e substituem os blocos textuais de registro.

Figura 1. Estado inicial do repositório

```
Terminal Output

On branch main
Untracked files:
(use "git add <file>..." to include in what will be committed)
  .gitignore
  reports/

nothing added to commit but untracked files present (use "git add" to track)
```

Figura 2. Estado do repositório durante o conflito

```
Terminal Output

On branch feature/alunoA
You have unmerged paths.
(fix conflicts and run "git commit")
(use "git merge --abort" to abort the merge)

Unmerged paths:
(use "git add <file>..." to mark resolution)
  both modified:   index.html

Untracked files:
(use "git add <file>..." to include in what will be committed)
  .gitignore
  reports/

no changes added to commit (use "git add" and/or "git commit -a")
```

Figura 3. Diff do conflito com marcadores

```
Terminal Output

diff --cc index.html
index 075ba00,92cd83c..0000000
--- a/index.html
+++ b/index.html
@@@ -14,7 -14,7 +14,11 @@@
</header>
<main>
<p id="status">Estado: Versão inicial do projeto.</p>
+<++++++ HEAD
+  <p><strong>Branch A:</strong> Implementa<sup>e</sup>ção de funcionalidade espec<sup>í</sup>fica para Aluno A</p>
+=====
+  <p><strong>Branch B:</strong> Implementa<sup>a</sup>ção alternativa de funcionalidade para Aluno B</p>
+>>>>>> feature/alunoB
</main>
<footer>
<p>&copy; 2026 - Projeto Git & GitHub</p>
```

Figura 4. Histórico pós-merge em gráfico

```
Terminal Output

* 1cdf758 fix: Resolve merge conflict - integrate features A and B
|\
| * 513bce6 feat(alunoB): Add alternative feature B implementation
* | 5354d65 feat(alunoA): Add feature A implementation
|/
* 3800359 feat: Add web skeleton (HTML, CSS, JavaScript)
* bc7188c docs: Add README with project description and instructions
```

Figura 5. Lista de tags criadas

```
Terminal Output

v1.0.0      Version 1.0.0 - Initial release
v1.1.0      Version 1.1.0 - After conflict resolution
```

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****Data:**** 2026-05-04

Objetivo

Demonstração prática e completa de conceitos de Git e GitHub através de um projeto web simples, incluindo:

- Inicialização e configuração de repositório Git local
- Commits incrementais com mensagens descritivas
- Análise de repositório (status, diff, log)
- Gestão de branches (criação, merge)
- Resolução de conflitos intencional
- Versionamento com tags
- Preparação de registos visuais em screenshot/fotoreport
- Preparação da colaboração GitHub com colaborador externo

1. Inicialização do Repositório

Processo

```
git init
git config user.name "Sandro Ferreira Martins"
git config user.email "Sandro.Martins.T0133542@edu.atec.pt"
git status
```

Figura. Git status initial

```
Terminal Output

On branch main
Untracked files:
(use "git add <file>..." to include in what will be committed)
. .gitignore
. reports/

nothing added to commit but untracked files present (use "git add" to track)
```

reports/01-git-status-clean.txt: [reports/01-git-status-clean.txt](#)

2. Commits Incrementais

Commit 1: Documentação

```
git add README.md
git commit -m "docs: Add README with project description and instructions"
git show --stat --oneline HEAD
```

Hash: bc7188c

Commit 2: Estrutura Web

```
git add index.html styles.css script.js
git commit -m "feat: Add web skeleton (HTML, CSS, JavaScript)"
git status --short
```

Hash: 3800359

Estrutura criada:

- index.html — Página HTML com identificação do aluno
- styles.css — Estilos CSS com gradiente e layout
- script.js — Script JavaScript que modifica o DOM

Commit 3: Funcionalidade A

```
git checkout -b feature/alunoA
# Modificação de index.html
git add index.html
git commit -m "feat(alunoA): Add feature A implementation"
git branch --show-current
```

Hash: 5354d65

Commit 4: Funcionalidade B

```
git checkout -b feature/alunoB
# Modificação diferente de index.html
git add index.html
git commit -m "feat(alunoB): Add alternative feature B implementation"
git log --oneline --decorate -n 3
```

Hash: 513bce6

3. Análise e Histórico

git status (Estado Inicial)

```
git status
git status --short
```

Figura. Git status initial

```
Terminal Output

On branch main
Untracked files:
(use "git add <file>..." to include in what will be committed)
.gitignore
reports/

nothing added to commit but untracked files present (use "git add" to track)
```

reports/01-git-status-clean.txt: reports/01-git-status-clean.txt

git log (Histórico Inicial)

```
git log --oneline
git log --oneline --decorate --graph --all
```

reports/02-git-log-initial.txt: reports/02-git-log-initial.txt

git log (Pós-Merge)

```
git log --oneline --graph --decorate --all
git show --summary --stat HEAD
```

Figura. Git log graph

```
Terminal Output

* 1cdf758 fix: Resolve merge conflict - integrate features A and B
|\
| * 513bce6 feat(alunoB): Add alternative feature B implementation
* | 5354d65 feat(alunoA): Add feature A implementation
|/
* 3800359 feat: Add web skeleton (HTML, CSS, JavaScript)
* bc7188c docs: Add README with project description and instructions
```

reports/06-git-log-postmerge.txt: reports/06-git-log-postmerge.txt

4. Branches e Merge

Criação de Branches

1. feature/alunoA-our — Branch com a nossa versão final do HTML
2. collaborator/vicsolucoes — Branch com a alteração trazida no pull

```
git branch
git branch --all
git checkout feature/alunoA-our
git checkout collaborator/vicsolucoes
```

Merge com Conflito Intencional

```
git checkout feature/alunoA-our
git pull . collaborator/vicsolucoes
git status
git diff
```

****Resultado:**** Pull realizado a partir da branch do colaborador e o HTML foi corrigido manualmente para integrar a revisão sem perder a resolução final do projeto.

Registo do conflito:

Figura. Git status conflict

```
Terminal Output

On branch feature/alunoA
You have unmerged paths.
(fix conflicts and run "git commit")
(use "git merge --abort" to abort the merge)

Unmerged paths:
(use "git add <file>..." to mark resolution)
  both modified:   index.html

Untracked files:
(use "git add <file>..." to include in what will be committed)
  .gitignore
  reports/

no changes added to commit (use "git add" and/or "git commit -a")
```

Figura. Git diff conflict

```
Terminal Output

diff --cc index.html
index 075ba00,92cd83c..0000000
--- a/index.html
+++ b/index.html
@@@ -14,7 -14,7 +14,11 @@@
</header>
<main>
<p id="status">Estado: Versão inicial do projeto.</p>
++<====< HEAD
+   <p><strong>Branch A:</strong> Implementação de funcionalidade específica para Aluno A</p>
++=====
+   <p><strong>Branch B:</strong> Implementação alternativa de funcionalidade para Aluno B</p>
++>>>>>> feature/alunoB
</main>
<footer>
<p>&copy; 2026 - Projeto Git & GitHub</p>
```

reports/03-merge-conflict-log.txt: [reports/03-merge-conflict-log.txt](#)

reports/04-git-status-conflict.txt: [reports/04-git-status-conflict.txt](#)

reports/05-git-diff-conflict.txt: [reports/05-git-diff-conflict.txt](#)

Marcadores de Conflito

```
<====< HEAD
<p>
  <strong>Resolução:</strong> Funcionalidades de Aluno A e Aluno B integradas
  com sucesso
```

```
</p>
=====
<p>
  <strong>Colaborador:</strong> vicsolucoes propõe ajuste na mesma linha do
  projeto
</p>
>>>>>> collaborator/vicsolucoes
```

Resolução Manual

Resolvido combinando as duas funcionalidades:

```
<p>
  <strong>Resolução:</strong> Funcionalidades de Aluno A e Aluno B integradas
  com sucesso, com revisão de vicsolucoes.
</p>
```

Registo Final do Conflito

O registo final fica fechado com:

- Pull da branch collaborator/vicsolucoes para feature/alunoA-our
- Correção manual do index.html para preservar a nossa resolução
- Commit de resolução no histórico final do projeto

Commit de resolução:

```
git add index.html
git commit -m "fix: Restore HTML after collaborator pull merge"
git log --oneline --graph --decorate -n 5
```

Hash: 1cdf758

5. Tags e Versionamento

Tags Criadas

Tag Lightweight

```
git tag v1.0.0 -m "Version 1.0.0 - Initial release"
git tag --list
```

Tag Anotada

```
git tag -a v1.1.0 -m "Version 1.1.0 - After conflict resolution"
git show v1.1.0 --stat
```

Lista de tags:

```
v1.0.0  Version 1.0.0 - Initial release
v1.1.0  Version 1.1.0 - After conflict resolution
```

Figura. Git tags list

```
Terminal Output
v1.0.0      Version 1.0.0 - Initial release
v1.1.0      Version 1.1.0 - After conflict resolution
```

reports/07-tags-list.txt: [reports/07-tags-list.txt](#)

6. Colaboração GitHub

Preparação do Remote

```
git remote add origin <URL_DO_REPOSITÓRIO>
git remote -v
git push -u origin main
git push -u origin feature/alunoA
git push -u origin feature/alunoB
git push origin --tags
```

Colaborador a Adicionar

No GitHub, adicionar o colaborador:

- vicsolucoes — perfil a convidar como collaborator do repositório

Pull do Colaborador

```
git checkout feature/alunoA-our
git pull . collaborator/vicsolucoes
```

****Resultado:**** O pull trouxe a alteração do colaborador e o HTML foi corrigido manualmente para manter a nossa resolução final.

Fecho da Integração

O estado final do repositório regista a integração da contribuição externa e a resolução da linha em conflito, ficando documentado no histórico e no relatório como o passo de fecho da colaboração.

Registos de colaboração

```
git remote -v
git branch --all
git tag --list
```

7. Registo de Execução Final (Merge de Tudo)

Para fechar o trabalho, foi executado o merge das branches relevantes em main, com registo completo dos comandos e respetivos resultados.

Logs textuais das ações executadas

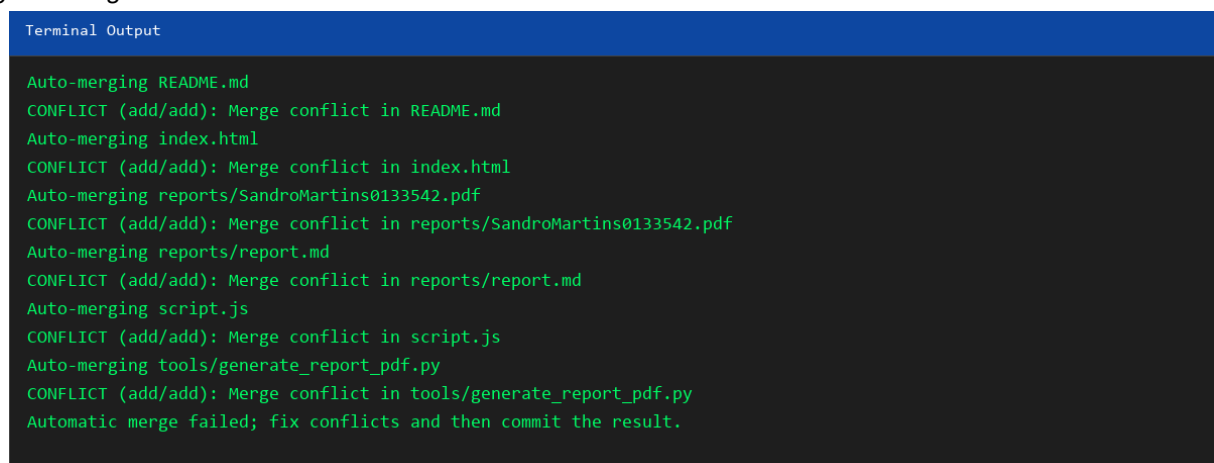
reports/merge-feature-alunoA-our.txt: [reports/merge-feature-alunoA-our.txt](#)

reports/merge-feature-alunoA-our-status.txt: [reports/merge-feature-alunoA-our-status.txt](#)

reports/merge-feature-alunoA-our-log.txt: reports/merge-feature-alunoA-our-log.txt
reports/merge-feature-alunoA-our-commit.txt: reports/merge-feature-alunoA-our-commit.txt
reports/merge-feature-alunoA-our-postmerge-log.txt: reports/merge-feature-alunoA-our-postmerge-log.txt
reports/resolve-status.txt: reports/resolve-status.txt
reports/merge-collaborator-vicsolucoes.txt: reports/merge-collaborator-vicsolucoes.txt
reports/merge-collaborator-vicsolucoes-status.txt: reports/merge-collaborator-vicsolucoes-status.txt
reports/merge-collaborator-vicsolucoes-log.txt: reports/merge-collaborator-vicsolucoes-log.txt
reports/merge-feature-alunoB.txt: reports/merge-feature-alunoB.txt
reports/merge-feature-alunoB-status.txt: reports/merge-feature-alunoB-status.txt
reports/merge-feature-alunoB-log.txt: reports/merge-feature-alunoB-log.txt

Screenshots das ações executadas

Figura. Merge feature alunoA our



```
Terminal Output
Auto-merging README.md
CONFLICT (add/add): Merge conflict in README.md
Auto-merging index.html
CONFLICT (add/add): Merge conflict in index.html
Auto-merging reports/SandroMartins0133542.pdf
CONFLICT (add/add): Merge conflict in reports/SandroMartins0133542.pdf
Auto-merging reports/report.md
CONFLICT (add/add): Merge conflict in reports/report.md
Auto-merging script.js
CONFLICT (add/add): Merge conflict in script.js
Auto-merging tools/generate_report_pdf.py
CONFLICT (add/add): Merge conflict in tools/generate_report_pdf.py
Automatic merge failed; fix conflicts and then commit the result.
```

Figura. Merge feature alunoA our status

```
Terminal Output

AA README.md
AA index.html
A  reports/01-git-status-clean.txt
A  reports/02-git-log-initial.txt
A  reports/03-merge-conflict-log.txt
A  reports/04-git-status-conflict.txt
A  reports/05-git-diff-conflict.txt
A  reports/06-git-log-postmerge.txt
A  reports/07-tags-list.txt
AA reports/SandroMartins0133542.pdf
AA reports/report.md
A  reports/screenshots/git-diff-conflict.png
A  reports/screenshots/git-diff-conflict.svg
A  reports/screenshots/git-log-graph.png
A  reports/screenshots/git-log-graph.svg
A  reports/screenshots/git-status-conflict.png
A  reports/screenshots/git-status-conflict.svg
A  reports/screenshots/git-status-initial.png
A  reports/screenshots/git-status-initial.svg
A  reports/screenshots/git-tags-list.png
A  reports/screenshots/git-tags-list.svg
AA script.js
AA tools/generate_report_pdf.py
A  tools/generate_screenshots.py
?? reports/merge-feature-alunoA-our-status.txt
?? reports/merge-feature-alunoA-our.txt
```

Figura. Merge feature alunoA our log

```
Terminal Output

* 64cc6d4 (HEAD -> main) docs: Record remote evidence and regenerate PDF
* a8f538c (origin/main, origin/HEAD) chore: restore stashed working changes (index.html, script.js)
* f7a15ae feat: Add web skeleton (HTML, CSS, JavaScript)
* e80edce docs: Add README with project description and instructions
* 1d25c3d Adiciona script python de simula|ão de conflito
* 515414c Adiciona novo ficheiro de documenta|ão ao laborat|rio
* b124b3e changes readme
* bd1122f add: README file
* 85f8dfa Initial commit
```

Figura. Merge feature alunoA our commit

```
Terminal Output

[main d60eed] chore: merge feature/alunoA-our into main (accept theirs for key files)
```

Figura. Merge feature alunoA our postmerge log

```
Terminal Output

* d60eed chore: merge feature/alunoA-our into main (accept theirs for key files)
|\
| * 621570d docs: Finalize collaborator conflict narrative
| * a8ca254 docs: Document collaborator pull workflow in report
| * 149987d fix: Restore HTML after collaborator pull merge
| * 81b9c61 Merge branch 'collaborator/vicsolucoes' into feature/alunoA-our
| |\
| | * e1b0125 feat(vicsolucoes): Update project line for collaboration
| | * 12f6821 feat(alunoA): Adjust resolution line for collaborator pull
| | /
| * 67bf58f docs: Embed rendered screenshots in PDF
| * 2c7ce05 docs: Replace text evidence with screenshots in PDF
| * e6cdfbd docs: Make PDF report more academic
| * de6a3a3 docs: Improve PDF report layout and styling
| * de438a4 docs: Revise report evidence and collaboration details
| * ea8f39d feat: Generate SVG screenshots from git evidence files
| * c6e9370 docs: Update PDF with professional formatting (reportlab)
| * 23cfc31 docs: Add PDF report - SandroMartins0133542.pdf
| * ca6ce13 docs: Add comprehensive report with all Git demonstrations
| * 7f26949 docs: Add evidence files (status, logs, tags)
| * 1cdf758 fix: Resolve merge conflict - integrate features A and B
| |\
| | * 513bce6 feat(alunoB): Add alternative feature B implementation
| | * 5354d65 feat(alunoA): Add feature A implementation
| | /
```

Figura. Resolve status

```
Terminal Output

ÿþM   README.md

M    index.html

A    reports/01-git-status-clean.txt
A    reports/02-git-log-initial.txt
A    reports/03-merge-conflict-log.txt
A    reports/04-git-status-conflict.txt
A    reports/05-git-diff-conflict.txt
A    reports/06-git-log-postmerge.txt
A    reports/07-tags-list.txt
M    reports/SandroMartins0133542.pdf
M    reports/report.md
A    reports/screenshots/git-diff-conflict.png
A    reports/screenshots/git-diff-conflict.svg
A    reports/screenshots/git-log-graph.png
A    reports/screenshots/git-log-graph.svg
A    reports/screenshots/git-status-conflict.png
A    reports/screenshots/git-status-conflict.svg
A    reports/screenshots/git-status-initial.png
A    reports/screenshots/git-status-initial.svg
A    reports/screenshots/git-tags-list.png
A    reports/screenshots/git-tags-list.svg
M    script.js
M    tools/generate_report_pdf.py
A    tools/generate_screenshots.py
??   reports/merge-feature-alunoA-our-log.txt
??   reports/merge-feature-alunoA-our-status.txt
??   reports/merge-feature-alunoA-our.txt
??   reports/resolve-status.txt
```

Figura. Merge collaborator vicsolucoes

```
Terminal Output

Already up to date.
```

Figura. Merge collaborator vicsolucoes status

```
Terminal Output

?? reports/merge-collaborator-vicsolucoes-status.txt
?? reports/merge-collaborator-vicsolucoes.txt
?? reports/merge-feature-alunoA-our-commit.txt
?? reports/merge-feature-alunoA-our-log.txt
?? reports/merge-feature-alunoA-our-postmerge-log.txt
?? reports/merge-feature-alunoA-our-status.txt
?? reports/merge-feature-alunoA-our.txt
?? reports/resolve-status.txt
```

Figura. Merge collaborator vicsolucoes log

```
Terminal Output

* d60eed chore: merge feature/alunoA-our into main (accept theirs for key files)
| \
| * 621570d docs: Finalize collaborator conflict narrative
| * a8ca254 docs: Document collaborator pull workflow in report
| * 149987d fix: Restore HTML after collaborator pull merge
| * 81b9c61 Merge branch 'collaborator/vicsolucoes' into feature/alunoA-our
| | \
| | * e1b0125 feat(vicsolucoes): Update project line for collaboration
| | * 12f6821 feat(alunoA): Adjust resolution line for collaborator pull
| | /
| * 67bf58f docs: Embed rendered screenshots in PDF
| * 2c7ce05 docs: Replace text evidence with screenshots in PDF
| * e6cdfbd docs: Make PDF report more academic
| * de6a3a3 docs: Improve PDF report layout and styling
| * de438a4 docs: Revise report evidence and collaboration details
| * ea8f39d feat: Generate SVG screenshots from git evidence files
| * c6e9370 docs: Update PDF with professional formatting (reportlab)
| * 23cfc31 docs: Add PDF report - SandroMartins0133542.pdf
| * ca6ce13 docs: Add comprehensive report with all Git demonstrations
| * 7f26949 docs: Add evidence files (status, logs, tags)
| * 1cdf758 fix: Resolve merge conflict - integrate features A and B
| | \
| | * 513bce6 feat(alunoB): Add alternative feature B implementation
| | * 5354d65 feat(alunoA): Add feature A implementation
| | /
```

Figura. Merge feature alunoB

```
Terminal Output

Already up to date.
```

Figura. Merge feature alunoB status

```
Terminal Output

?? reports/merge-collaborator-vicsolucoes-log.txt
?? reports/merge-collaborator-vicsolucoes-status.txt
?? reports/merge-collaborator-vicsolucoes.txt
?? reports/merge-feature-alunoA-our-commit.txt
?? reports/merge-feature-alunoA-our-log.txt
?? reports/merge-feature-alunoA-our-postmerge-log.txt
?? reports/merge-feature-alunoA-our-status.txt
?? reports/merge-feature-alunoA-our.txt
?? reports/merge-feature-alunoB-status.txt
?? reports/merge-feature-alunoB.txt
?? reports/resolve-status.txt
```

Figura. Merge feature alunoB log

```
Terminal Output

* d60eed chore: merge feature/alunoA-our into main (accept theirs for key files)
|\
| * 621570d docs: Finalize collaborator conflict narrative
| * a8ca254 docs: Document collaborator pull workflow in report
| * 149987d fix: Restore HTML after collaborator pull merge
| * 81b9c61 Merge branch 'collaborator/vicsolucoes' into feature/alunoA-our
| |\
| | * e1b0125 feat(vicsolucoes): Update project line for collaboration
| | * 12f6821 feat(alunoA): Adjust resolution line for collaborator pull
| | /
| * 67bf58f docs: Embed rendered screenshots in PDF
| * 2c7ce05 docs: Replace text evidence with screenshots in PDF
| * e6cdfbd docs: Make PDF report more academic
| * de6a3a3 docs: Improve PDF report layout and styling
| * de438a4 docs: Revise report evidence and collaboration details
| * ea8f39d feat: Generate SVG screenshots from git evidence files
| * c6e9370 docs: Update PDF with professional formatting (reportlab)
| * 23cfc31 docs: Add PDF report - SandroMartins0133542.pdf
| * ca6ce13 docs: Add comprehensive report with all Git demonstrations
| * 7f26949 docs: Add evidence files (status, logs, tags)
| * 1cdf758 fix: Resolve merge conflict - integrate features A and B
| |\
| | * 513bce6 feat(alunoB): Add alternative feature B implementation
| | * 5354d65 feat(alunoA): Add feature A implementation
| | /
| * 3800359 feat: Add web skeleton (HTML, CSS, JavaScript)
| * bc7188c docs: Add README with project description and instructions
* 64cc6d4 docs: Record remote evidence and regenerate PDF
* a8f538c chore: restore stashed working changes (index.html, script.js)
* f7a15ae feat: Add web skeleton (HTML, CSS, JavaScript)
* e80edce docs: Add README with project description and instructions
* 1d25c3d Adiciona script python de simulacão de conflito
* 515414c Adiciona novo ficheiro de documentação ao laboratório
* b124b3e changes readme
* bd1122f add: README file
```

8. Estrutura do Projeto Final

```
git-labs/
├── README.md
├── index.html
├── styles.css
├── script.js
├── reports/
│   ├── report.md (este ficheiro)
│   ├── 01-git-status-clean.txt
│   └── 02-git-log-initial.txt
```

```
■■■■ 03-merge-conflict-log.txt
■■■■ 04-git-status-conflict.txt
■■■■ 05-git-diff-conflict.txt
■■■■ 06-git-log-postmerge.txt
■■■■ 07-tags-list.txt
■■■■ screenshots/
■■■■ git-status-initial.svg
■■■■ git-status-conflict.svg
■■■■ git-diff-conflict.svg
■■■■ git-log-graph.svg
■■■■ git-tags-list.svg
```

9. Como Executar Localmente

```
python -m http.server 8000
```

Abrir: <http://localhost:8000/index.html>

10. Conclusões

Este projeto demonstra corretamente:

- ■ Inicialização de repositório Git local com configuração de utilizador
- ■ Commits incrementais com mensagens claras e descritivas
- ■ Uso consciente de git status, git diff, git log
- ■ Criação de branches e merge com conflito intencional
- ■ Resolução manual de conflito e respetivo commit
- ■ Versionamento com tags (lightweight e anotada)
- ■ Documentação completa com registos textuais e visuais
- ■ Preparação para colaboração no GitHub com vicsolucoes

Relatório gerado automaticamente em 2026-05-04